

Diffusion model:

$$\mathbf{x}_k = a_k \mathbf{x}_0 + \sigma_k \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

on frequency, it is equivalently with:

$$\mathbf{x}_k(\omega) = a_k \mathbf{x}_0(\omega) + \sigma_k \boldsymbol{\eta}(\omega)$$

with $\boldsymbol{\eta}(\omega)$ is spectrally flat Gaussian noise. Signal-to-noise ratio (SNR) at noise level k is:

$$s_k(\omega) = \frac{a_k^2}{\sigma_k^2} P(\omega),$$

with $P(\omega) = \mathbb{E} \left[|\mathbf{x}_0(\omega)|^2 \right]$ is the power of frequency ω .

0.1 Motivation

- Normal diffusion models employ same noise scheduler for all frequencies of the signal. Therefore, all frequencies receive same amount of noise over the diffusion process.
- However, different frequencies have different signal power: low frequencies (large structure) power are significantly larger than high-frequencies power (finer details), and the power decays with frequency by orders of magnitude.
- It implies that the SNR $s_k(\omega)$ of high frequencies decay faster, and before low frequency in the forward process. During reverse process, the low frequencies are generated in the high-to-mid noise levels, while high frequencies are generated in mid-to-low noise levels.
- Current VDMs (Wan 2.1, LTX-Video) use inference noise scheduler that skew towards the high-to-mid noise levels, making the generation focus significantly to generate low frequencies while overlooking high frequencies.
- We argue that instead of using same noise scheduler for all frequencies, we can employ frequency-dependent noise schedulers. More concretely, the noise scheduler for low frequencies will pay attention to the high-to-mid noise, while high frequencies will pay more attention to the mid-to-low noise.
- However, the model only take 1 noise level value. Therefore, we need to compute a global noise level for all frequencies.

0.2 Derivation

Let $\kappa_k(\omega)$ is the effective timestep assigned to frequency ω when the global sampling timestep is k . The goal is to choose $\kappa_k(\omega)$ so that every frequency has approximately the same SNR at global step k . That is, impose

$$s_{\kappa_k}(\omega) = \rho_k,$$

where ρ_k is the target SNR value at global step k .

Diffusion Model Give $a_k^2 + \sigma_k^2 = 1$, we have

$$\frac{1 - \sigma_{\kappa_k}^2(\omega)}{\sigma_{\kappa_k}^2(\omega)} = \frac{\rho_k}{P(\omega)}$$

We choose

$$\begin{aligned} \rho_k &= \frac{1}{|\Omega|} \sum_{\omega \in \Omega} s_k(\omega) \\ &= \frac{1}{|\Omega|} \sum_{\omega \in \Omega} \frac{a_k^2}{\sigma_k^2} P(\omega) \\ &= \frac{a_k^2}{\sigma_k^2} \bar{P} \end{aligned}$$

$$\begin{aligned} \sigma_{\kappa_k}^2(\omega) &= \frac{P(\omega)}{\rho_k + P(\omega)} \\ &= \frac{\sigma_k^2 P(\omega)}{a_k^2 \bar{P} + \sigma_k^2 P(\omega)} \end{aligned}$$

It means we need to choose global timesteps first, then compute the frequency-dependent timesteps based on its own power.

Flow matching Given $a_k = 1 - \sigma_k$:

$$\left(\frac{1 - \sigma_{\kappa_k}(\omega)}{\sigma_{\kappa_k}(\omega)} \right)^2 = \frac{\rho_k}{P(\omega)}$$

Therefore

$$\sigma_{\kappa_k}(\omega) = \frac{1}{1 + \sqrt{\frac{\rho_k}{P(\omega)}}}$$

We choose

$$\begin{aligned}
\rho_k &= \frac{1}{|\Omega|} \sum_{\omega \in \Omega} s_k(\omega) \\
&= \frac{1}{|\Omega|} \sum_{\omega \in \Omega} \frac{(1 - \sigma_k)^2}{\sigma_k^2} P(\omega) \\
&= \frac{(1 - \sigma_k)^2}{\sigma_k^2} \bar{P}
\end{aligned}$$

$$\begin{aligned}
\sigma_{\kappa_k}(\omega) &= \frac{1}{1 + \sqrt{\frac{\rho_k}{P(\omega)}}} \\
&= \frac{1}{1 + \frac{1 - \sigma_k}{\sigma_k} \sqrt{\frac{\bar{P}}{P(\omega)}}} \\
&= \frac{\sigma_k}{\sigma_k + (1 - \sigma_k) \sqrt{\frac{\bar{P}}{P(\omega)}}}
\end{aligned}$$

It means we need to choose global timesteps first, then compute the frequency-dependent timesteps based on its own power.

0.3 Softer version

Diffusion Models

$$\begin{aligned}
\rho_{\kappa_k}(\omega) &= s_k(\omega)^{1-\gamma} \rho_k^\gamma \\
&= \frac{a_k^{2(1-\gamma)}}{\sigma_k^{2(1-\gamma)}} P(\omega)^{(1-\gamma)} \frac{a_k^{2\gamma}}{\sigma_k^{2\gamma}} \bar{P}^\gamma \\
&= \frac{a_k^2}{\sigma_k^2} P(\omega)^{(1-\gamma)} \bar{P}^\gamma
\end{aligned}$$

$$\begin{aligned}
\sigma_{\kappa_k}^2(\omega) &= \frac{P(\omega)}{\rho_k + P(\omega)} \\
&= \frac{P(\omega)}{\frac{a_k^2}{\sigma_k^2} P(\omega)^{(1-\gamma)} \bar{P}^\gamma + P(\omega)} \\
&= \frac{\sigma_k^2}{\sigma_k^2 + a_k^2 \left(\frac{\bar{P}}{P(\omega)} \right)^\gamma}
\end{aligned}$$

Flow matching

$$\begin{aligned}
\rho_{\kappa_k}(\omega) &= s_k(\omega)^{1-\gamma} \rho_k^\gamma \\
&= \frac{(1-\sigma_k)^{2(1-\gamma)}}{\sigma_k^{2(1-\gamma)}} P(\omega)^{(1-\gamma)} \frac{(1-\sigma_k)^{2\gamma}}{\sigma_k^{2\gamma}} \bar{P}^\gamma \\
&= \frac{(1-\sigma_k)^2}{\sigma_k^2} P(\omega)^{(1-\gamma)} \bar{P}^\gamma
\end{aligned}$$

$$\begin{aligned}
\sigma_{\kappa_k}(\omega) &= \frac{1}{1 + \sqrt{\frac{\rho_{\kappa_k}(\omega)}{P(\omega)}}} \\
&= \frac{1}{1 + \frac{1-\sigma_k}{\sigma_k} \sqrt{\frac{P(\omega)^{(1-\gamma)} \bar{P}^\gamma}{P(\omega)}}} \\
&= \frac{1}{1 + \frac{1-\sigma_k}{\sigma_k} \sqrt{\left(\frac{\bar{P}}{P(\omega)}\right)^\gamma}} \\
&= \frac{\sigma_k}{\sigma_k + (1-\sigma_k) \sqrt{\left(\frac{\bar{P}}{P(\omega)}\right)^\gamma}}
\end{aligned}$$